LAB EXPERIMENT

13. Perform Affine Transformation on the given image using python and Open CV

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread('image.jpg')
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

rows, cols = img.shape[:2]

src = np.float32([[50, 50], [200, 50], [50, 200]])
dst = np.float32([[10, 100], [200, 50], [100, 250]])

M = cv2.getAffineTransform(src, dst)
result = cv2.warpAffine(img, M, (cols, rows))

plt.subplot(1, 2, 1)
plt.imshow(img)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(result)
plt.title('Affine Transformation')
plt.axis('off')
plt.show()
```

